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UNIVERSITI TUNKU ABDUL RAHMAN

UGEA3183 DIGITAL SIGNAL PROCESSING

LAB 1 REPORT

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Programme : Bachelor of Engineering (Honours) Electronic Engineering
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Introduction

This laboratory session introduces the uses of Texas Instruments TMS320C6713 DSK, a development platform designed for real time digital signal processing applications. This session focused on familiarizing the hardware features of the DSP board and the uses of CCS (Code Compiler Studio). CCS is an integrated development environment for program stage including development, compilation, and debugging.

In this lab, practical experience such as creates and learn how to run simple DSP programs, generate signals, and observe the effects of sampling in digital systems. By going through the lookup tables and audio interfaces, a sinusoidal signal is generated and analysed using software analysing tools such as spectrum analyser and oscilloscopes. The whole procedures emphasizing the basic workflow of DSP system development and provides hands-on experience with Real-time signal processing implementation.

Results

Discussions

In Assignment 1, sine waves are generated using different sampling frequencies of 32kHz, 16kHz and 8kHz. The results show that sampling frequency directly affect the characteristic and quality of output signal produced. The relationship between output frequency, sampling frequency and number of samples can be explained by the formula (www.ni.com, n.d.):

$$Output\ frequency, f_{OUT} = \frac{Sample\ frequency, f_s}{Samples\ per\ cycle, N}$$

In the program, a sine wave is generated using a lookup table that stores several sample values representing one full complete cycle of a sine signal. The DSP outputs these sample one by one at the sampling rate f_s. After all N samples is sent (where N=8 is set in assignment 1), the program immediately returns to the first sample and repeating the cycle continuously. Since one cycle requires N samples, and the systems produces f_s samples per second, the number of cycles produced in one second is equal to the sampling rate divided by the number of samples per cycle. Therefore, increasing in sampling frequency will cause output frequency to increase while increasing in number of samples per cycle decreases the output frequency as more samples are needed to complete one cycle.

As the sample frequency is set to 32kHz, the waveform appears to be smooth in time domain because more samples were taken in one second, allowing the signal to be better in presenting the sinusoidal shape. As observed in figure X, there are more sine wave accumulate

within the time zone of 10ms, representing high frequency output. In the frequency domain, the waveform showing a clear peak of 4kHz with minimal distortion. This result confirms the equation above and showing that the signal was generated correctly as 32000Hz/8 = 4000Hz. The small difference between marker value of 4005Hz with the actual 4000Hz is due to the frequency resolution of the spectrum analyser, which is 11Hz. Because of this resolution limitations, the analyser marker may show value of slightly above or below the actual output frequency. We can prove this theory by observing the next output frequency, which is 16kHz and 8kHz. As observed in figure X when the sampling frequency is 16kHz, the number of sine wave accumulate within 10ms time zone is decreased, representing lower frequency output than in figure X. While for the spectrum analyser in figure X, the waveform showing a peak of 2kHz. As for the last part in assignment 1, as we keep lowering the sampling frequency to 8kHz, the decreased number of output frequencies waveform in 10ms is observed in figure X while a lower peak frequency of 1kHz is observed through spectrum analyser in figure X. These results demonstrate the importance of sampling frequency in digital signal processing, where higher sampling rates will produce more accurate and higher analogue output signals. As for the audio output signal through the earphones, different pitch and tones were heard depending on the sampling frequency used. When the sampling frequency is 32kHz, a high pitch note was clearly heard which corresponded to output frequency of 4kHz. This is reasonable as 4kHz of frequency falls within the audible range of human ears therefore can be easily perceived. The pitch is getting lower as we decrease the sampling frequency from 32kHz to 8kHz, indicates that strength of the pitch is directly proportional to the output frequency (Online Learning College, 2022), while also proving that the DSP board successfully generated the sinusoidal signals according to the sampling frequency used.

In assignment 2, the effect of changing the number of samples in the lookup table was investigated. Two configurations were tested: a sampling frequency of 32 kHz with 10 samples per cycle, and a sampling frequency of 8 kHz with 12 samples per cycle. For the first configuration, the waveform appeared relatively smooth due to high sampling rate. In figure X, we can observe that the oscilloscope showing more sine wave in time frame of 10ms while in spectrum analyser of figure X, we saw a peak of 3.2kHz which perfectly aligned with the relationship between number of sample and output frequency. In another configuration of 8kHz with 12 samples per cycle, there is lesser waveform in time frame of 10ms observed from oscilloscope in figure X while the spectrum analyser in figure X having a peak output frequency of 667Hz.

Overall, the experiment demonstrated that both sampling frequency and the number of samples per cycle influence the quality and strength of the generated output digital signal. Higher sampling frequencies improve signal continuity over time and higher output frequency, while increasing the number of samples per cycle improves the approximation of the sinusoidal waveform shape but lower output frequency (Baker, 2003).

Conclusion

In conclusion, Lab 1 successfully fulfilled its objective in providing an exposure to the TM320C6713 DSK development environment using the Code Composer Studio development platform. Such objective is achieved through the implementation of sine wave generation which demonstrated the relationship between the sampling and output frequency, in addition to the number of samples per unit cycle during digital signal processing and analysis.

In reference to the first assignment, the sampling frequency has been proven to have a direct effect on the output signal quality and waveform quality. This is shown by a higher sampling frequency has resulted in a higher output frequency and a smoother time-domain waveforms. The corresponding frequency-domain analysis also displayed peaks that display a similar value to the theoretical values. Minor difference between the theoretical and expected values are observed nevertheless, however these differences are classifiable under the problems of resolution limitations of the spectrum analyser and external noises. Such claim is supported further by the audible differences in the pitch.

For assignment 2, the increment of lookup table size improved the approximation of the sine wave by changing the number of samples per cycle. However as a side effect, the output frequency is reduced for a fixed sampling rate, indicating a potential trade-off between the resolution of the waveform and the frequency of the signal generated during digital waveform generation.

In summary, this experiment allows a clearer understanding of signal generation in digital spectrum in addition to how sampling affect signal behaviour and the real-life implementation of digital signal processing programming using TM320C6713 DSK development environment. Thus emphasizing the importance of utilizing and choosing the correct parameters during signal sampling to obtain the expected characteristics in future applications

References

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